On using the Hot Plate Test to characterize the Analgesic Profiles of three Novel Chemical Entities that inhibit the Glycine Transporter 2 in ann Acute Pain Mouse Model

Oishey Hossain

SID: 490432945

Results

We observed a significant difference across the 'pain'-condition; F (4, 45) = 685, p < .0001. Morphine-(30mg/kg) yielded a higher mean-latency (M = 23.02, SD = 1.23) than VEH (M = 5, SD = 0.83); p < .0001. However, Figure 1. and Table 1. shows NCE1's null-effect as all doses yielded mean-latencies similar to VEH, p > .05.

We observed a significant difference across the 'mouse'-condition; F (4, 45) = 501.5, p < .0001. Table 2. reports the significant, dose-dependent increases in mean-latencies for NCE2 (3-, 10-, 30-mg/kg), p < .0001. Figure 2. shows all NCE2-doses generated mean-latencies shorter than Morphine-(30mg/kg), p < .0001, while only NCE2-(3 mg/kg) was comparable to VEH, p > .05.

We observed a significant difference across the 'drug'-condition; F (4, 45) = 543.5, p < .0001. Table 3. reports the significant, dose-dependent increases in mean-latencies for NCE2, p < .0001. Figure 3. Shows all NCE3 doses yielded latencies greater than VEH, p < .0001, while only NCE3-(10 mg/kg) yielded a mean-latency higher than Morphine-(30mg/kg), p < .0001. Figure 3. also shows although, NCE3-(1-mg/kg) yielded a mean-latency lower than Morphine-(30-mg/kg), p < .0001, NCE2-(3-mg/kg) yielded a mean-latency identical to Morphine-(30-mg/kg), p >.005.

Table 1.

Descriptive Statistics for the 'Pain' Experimental Condition

Drug-Dose	N	Mean Latency	Std. Dev.	Skewness	
				Statistic	Std. Error
Control/VEH	10	5.000	.8250	-.6262	.2609
NCE1-(10-mg/kg)	10	4.825	.6977	-.1173	.2206
NCE1-(30-mg/kg)	10	4.950	1.066	.2907	.3371
NCE1-(100-mg/kg)	10	5.275	.9534	.6253	.3015
Morphine-(30-mg/kg)	110	23.02	1.234	.3841	.3903

Note. The table displays the descriptive statistics for the experiment that tested three doses of NCE1: 10 mg/kg, 30 mg/kg, 100 mg/kg. The differences in mean latencies generated by all NCE1 doses was similar to that of VEH. There was no test effect for NCE1.

Table 2.

Descriptive Statistics for the 'Mouse' Experimental Condition

Drug-Dose	N	Mean Latency	Std. Dev.	Skewness	
				Statistic	Std. Error
Control/VEH	10	5.500	.8580	-.7216	.2713
NCE2-(3-mg/kg)	10	5.600	.7379	.0934	.2333
NCE2-(10-mg/kg)	10	10.05	1.092	.5621	.3452
NCE2-(30-mg/kg)	10	17.08	1.302	.0640	.4117
Morphine-(30-mg/kg)	10	22.85	1.254	-.2822	.3965

Note. The table displays the descriptive statistics for the experiment that tested three doses of NCE2: 3 mg/kg, 10 mg/kg, 30 mg/kg. We found a dose-dependent effect for NCE2 with the lowest 3 mg/kg dose being comparable to VEH. All doses produced significantly shorter latencies compared to 30 mg/kg of morphine.

Table 3.

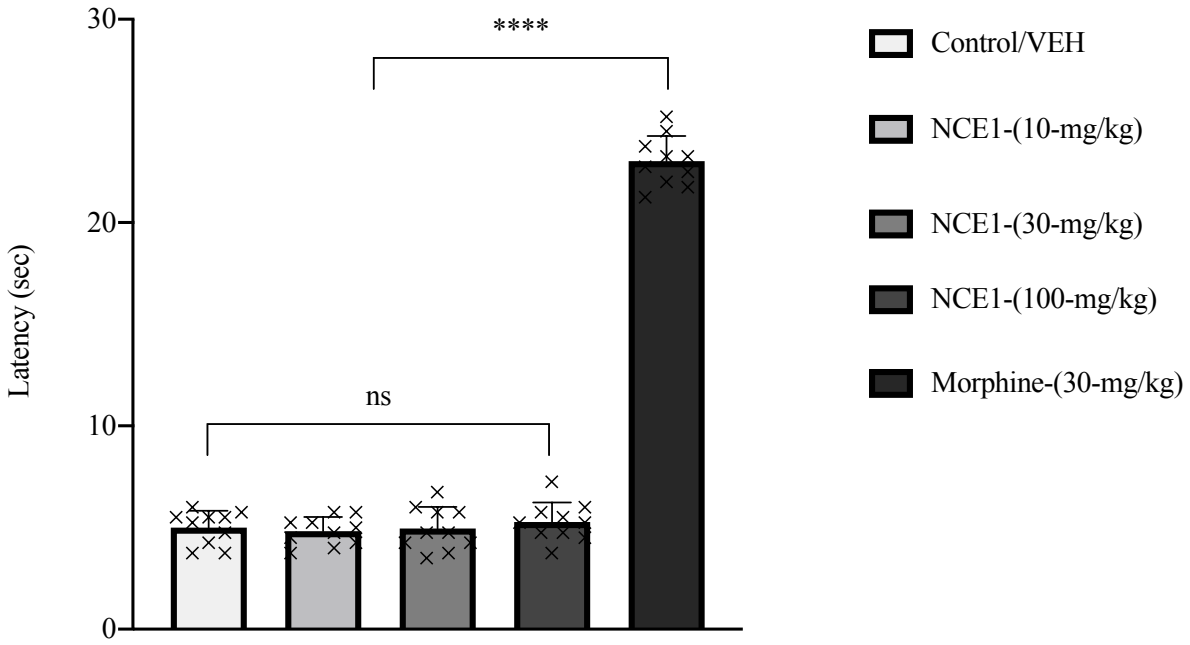
Descriptive Statistics for the 'Drug' Experimental Condition

Drug-Dose	N	Mean Latency	Std. Dev.	Skewness	
				Statistic	Std. Error
Control/VEH	10	5.375	.9373	-.3321	.2964
NCE3-(1-mg/kg)	10	11.08	1.586	.9246	.5015
NCE3-(3-mg/kg)	10	22.15	1.237	-.0652	.3912
NCE3-(10-mg/kg)	10	26.65	1.150	-.1668	.3636
Morphine-(30-mg/kg)	10	22.05	.9916	-.8525	.3136

Note. The table displays the descriptive statistics for the experiment that tested the three doses of NCE3: 1 mg/kg, 3 mg/kg, 10 mg/kg. We found a dose-dependent effect for NCE3, where the 3 mg/kg dose yielded a similar mean latency to 30 mg/kg of morphine. All doses yielded significantly greater latencies compared to VEH, while the highest dose of 10 mg/kg yielded a latency significantly greater than 30 mg/kg of morphine.

Figure 1.

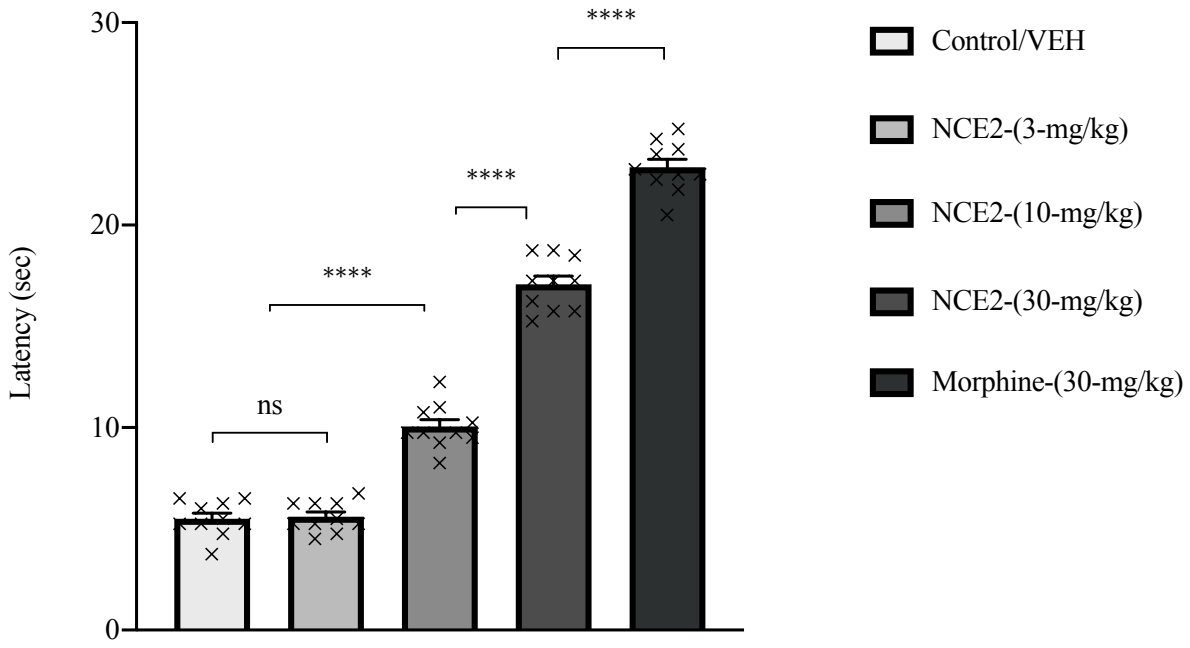
Comparison of mean latencies yielded by different NCE1 doses against VEH and Morphine



Note. The graph shows no differences in mean latencies across VEH and the 10 mg/kg, 30 mg/kg, and 100 mg/kg doses of NCE1. 30 mg/kg of morphine yielded a significantly higher mean latency than the latency yielded by all NCE1 doses and VEH. Plotted crosses represent the data obtained from a single subject. 'ns' refers to a p-value that is insignificant, i.e., $p > .05$. '****' is reflective of significance where $p < .0001$.

Figure 2.

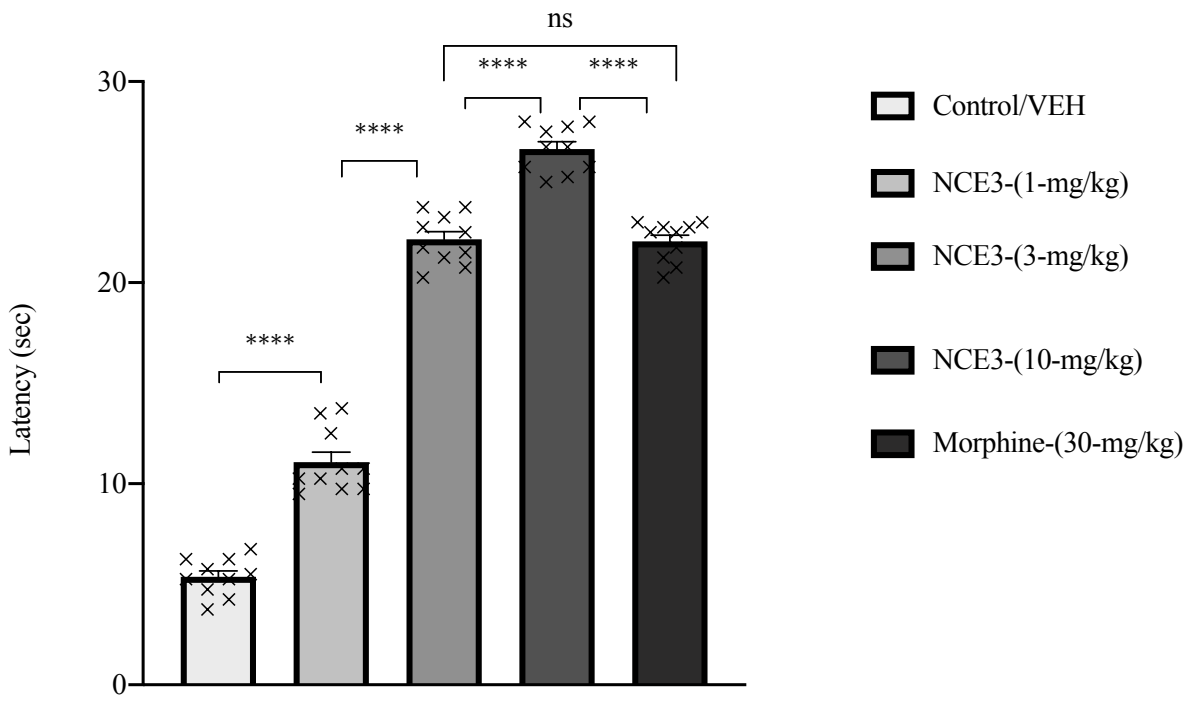
Comparison of mean latencies yielded by different NCE2 doses against VEH and Morphine



Note. The graph shows the mean-latencies yielded by VEH, 30 mg/kg of morphine, and three doses of NCE2: 3 mg/kg, 10 mg/kg, and 30 mg/kg. The lowest NCE2 dose of 3 mg/kg yielded a mean-latency similar to VEH's mean-latency. Although NCE2 shows an increasing dose-dependent response, all doses produced significantly shorter latencies than 30 mg/kg of morphine. Plotted crosses represent the data obtained from a single subject. 'ns' refers to a p-value that is insignificant, i.e., $p > .05$. '****' is reflective of significance where $p < .0001$.

Figure 3.

Comparison of mean latencies yielded by different NCE3 doses against VEH and Morphine



Note. The graph shows the mean-latencies yielded by VEH, 30 mg/kg of morphine, and three doses of NCE3: 1 mg/kg, 3 mg/kg, and 10 mg/kg. All doses yielded significantly higher mean latencies than control. The largest 10 mg/kg dose of NCE3 produced a higher mean latency than 30 mg/kg of morphine. The mean latency yielded by 3 mg/kg of NCE3 was comparable to 30 mg/kg of morphine. Plotted crosses represent the data obtained from a single subject. 'ns' refers to a p-value that is insignificant, i.e., $p > .05$. '****' is reflective of significance where $p < .0001$.

Discussion

A compromised supraspinal-inhibition of noxious-input at the superficial dorsal-horn fosters neuropathic-pain symptoms (e.g., hyperalgesia/allodynia) (Zeilhofer et al., 2017). Following i.p. injections of the Glycine-Transporter-2 inhibitor, Compound-33, Mostyn et al. 2019 showed a significantly increased von-Frey threshold over 90-minutes, compared to controls, reflective of allodynia-reversal, in a chronic-nerve-constriction model. Inhibition of GlyT2 on presynaptic-terminals of Lamina-3 glycinergic-interneurons, and descending glycinergic-projections from the rostroventral-medulla, likely restore the feed-forward inhibition of the excitatory-interneurons in Lamina-2 that are responsible for transmitting Aß-input from deeper laminae to the pain-projecting neurons of Lamina-1 in mouse models of allodynia (Vandenberg et al., 2014). Our NCEs may engage in similar mechanisms.

Currently, treatment comprises antidepressants, antiepileptics, and opioids. Antidepressants inhibit norepinephrine-reuptake in pain-modulating, descending locus-coeruleus projections to directly inhibit acute noxious inputs while indirectly increasing inhibitory GABA-interneuron activity in neuropathic-pain states via activating excitatory α2ARs on cholinergic-interneurons (Obata, 2017). Gabapentinoids may promote the latter by selectively inhibiting LC GABA-ergic activity while down-regulating N-type calcium channels on C-fibres to attenuate hyperalgesia (Sidhu and Sadhotra, 2016). While inhibiting glutamate and Substance-P release via blocking calcium-channels, opioids also increase pain-thresholds by hyper-polarising post-synaptic cells (Cohen and Preuss, 2019). As misuse/overdose limits opioid-efficacy and higher doses of antidepressants/antiepileptics for analgesia correspond to vast side effects, the novel GlyT2-inhibitors may offer more selective treatment for neuropathic-pain due to their restricted expression within the pain-pathways (Fornasari, 2017).

Therefore, glycine's role in pain-transmission substantiates our NCEs' preliminary-testing in a hot-plate test where responses are supraspinally-organized (Gunn et al., 2011). The

observed effects for NCE-2/NCE-3 likely derive from increased glycinergic-activity within descending-modulatory pathways similar to ALX1393-treated mice's observation where subsequent strychnine-administration reversed any thermal-analgesia observed on the hot-plate (Haranishi et al., 2010). Theoretically, an increase in glycinergic-transmission following GlyT2-inhibition precedes presynaptic-glycine-depletion, which concomitantly increases pain-transmission (Vandenberg et al., 2014). Different latencies yielded by NCE-2 and NCE-3 may stem from differences in transport-recovery and toxicity. Findings presented by Mostyn et al. 2019 substantiate this explanation. Following the significantly higher von-Frey-threshold after i.p. injections of Compound-33-(30-mg/kg) compared to ORG25543-(30mg/kg) 's insignificant-effect, electrophysiology-studies revealed that transport recovered within 25-minutes for the former while the latter only showed a 35% recovery at 30 minutes (Mostyn et al. 2019). In a formalin-induced pain-model, Mingorance-Le-Meur et al. 2013 showed a dose-dependent decrease in the late phase for ORG25543-(0.06-20mg/kg) with no significant response in the first-phase, while Haranishi et al. 2010 showed a dose-dependent increase for ALX1393-(4-40mg/kg) in both phases. While the pain-benefits for the former GlyT2-inhibitor may be apparent in the presence of central-nociceptive sensitization due to persistent pain, the latter may also effectively reduce pain-responses to acute input conveyed via peripheral A-delta and C-fibres (Lopez-Cano et al., 2017). This analysis is vital to understand the null-effects yielded by NCE1 for which a formalin-induced pain model, similar to Mingorance-Le-Meur et al. 2013 for ORG25543, may unmask the potential pain-benefits associated.

Extensive research on ORG25543 and ALX1393 highlight the questionability of targetting GlyT2. Takahashi et al. 2015 and Haranishi et al. 2010 only showed decreased pain-behavior for ALX1393 in a formalin-induced pain model following central-administration. Mingorance-Le-Meur et al. 2013 showed that following systemic administration, the ratio of ALX1393 that crossed the BBB after 60 minutes was .05. Thus, null-effects of NCE1 could

also stem from BBB-impermeability. As ALX1393 also partially inhibits GlyT1, the effective-doses of systemically-administered ALX1393 are associated with a wide range of adverse side-effects, including tremors, seizures, and respiratory-depression (Zeilhofer et al., 2017). Thus, future studies should include a rotarod-test to account for possible confounds introduced by motor coordination differences in our study.

Moreover, although glycinergic-activity initially increases, in-vitro studies show that ORG-inhibition eventually decreases eIPSCs to an extent reflective of hindered synaptic-signaling due to completely dysfunctional presynaptic-glycine-recycling (Winters et al., 2018). Thus, although NCE-2/NCE-3 increased latencies, their long-term efficacy remains inconclusive. Ultimately, our NCEs' promotion into clinical-trials will depend on their chronic-pain-model performance as glycines' role is more relevant to neuropathic-pain than acute-pain. We recommend evaluating NCEs' capacity to reverse allodynia by comparing von-Frey thresholds following GlyT2-inhibition after partial-nerve-ligation against the performance of Compound-33 (Mostyn et al., 2019). Furthermore, Santenna et al. 2019 demonstrate that the specific behavioral-analgesiometer engaged will constrain the time of maximum-response due to differential organization at supraspinal and spinal levels. A tail-flick-test may be more sensitive to the NCEs' effects as GlyT2-inhibitors primarily reduce pain-behavior by reducing excitability within pain-processing spinal-circuits (Winters et al., 2018). Asides from evaluating our NCEs' in a formalin-induced pain model to discriminate between neurogenic and inflammatory responses, prospective acute-pain models should compare latencies across the hot plate test and the tail-flick test to distinguish between spinally-organized and supra-spinally organized responses (Santenna et al., 2019).

The current study uses the hot plate test to enhance the foundation of GlyT2-inhibitors in the treatment of neuropathic pain. An effect-size yielded by NCE3-(3-mg/kg), comparable with Morphine-(30-mg/kg), encourages future-studies to engage different behavioral-assays in

chronic and acute models to elucidate the NCE3s' broader effects on the dynamic pain experience. The research line will yield a better understanding of GlyT2-mediated inhibition of pain-transmission, which will, in turn, promote the use of these GlyT2-inhibitors in clinical settings.

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